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JAIPUR NATIONAL UNIVERSITY

Semester/Year End Examination Dec.2024

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Computer Science) / B.Tech. CSE (Hons) in Artificial Intelligence and Machine Learning / B.Tech. CSE (Hons) in Cyber Security
Semester/Year:	3
Paper Code:	BTECSCO304T22
Title of Paper:	Analog Electronics Circuits
Time: 3 Hours	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section-A and Section-B. Section-A consists of seven questions and Section-B consists of six questions. Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

Question
Number

Question

Marks
Allotted

SECTION - A

- | | | |
|----------------|--|---|
| Q.1 | What is semiconductor according to the energy band gap theory? | 4 |
| Q.2 | Differentiate drift and diffusion current. | 4 |
| Q.3 | What is the relation between current gain α and β ? | 4 |
| Q.4 | What is mobility? Define. | 4 |
| Q.5 | What is load line in transistor? | 4 |
| Q.6 | What is Oscillator? Define Barkhausen criteria for Oscillation? | 4 |

Q.7 Explain FET as VVR.

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SECTION - B

- Q.1 Briefly explain CE transistor configuration with circuit diagram and also explain the "Transistor as an Amplifier". 10
- Q.2 Calculate the value of the resistors R_C and R_B on a fixed bias circuit using the following data $I_{CO} = 9 \text{ mA}$, $V_{CE} = 4.4 \text{ volts}$, $\beta = 110$, $V_{BE} = 0.7 \text{ V}$ and $V_{CC} = 8 \text{ V}$. 10
- Q.3 Draw and analysis of Common Emitter circuit using h-parameter and find A_i and R_i . 10
- Q.4 Draw and explain the two stage amplifier with 'Darlington connection'. Give advantage of this circuit. 10
- Q.5 Explain the working of Colpitts oscillator and find the frequency of oscillation. 10
- Q.6 Explain Op-Amp as a differential amplifier? Also define CMRR and Slew rate. 10